

From Equations to Code

Implementing E(n)-Equivariant Topological Neural Networks in TopoBench

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This technical report explains the design and implementation of an E(n)-Equivariant Topological Neural Network (ETNN) inside TopoBench. The account follows the implementation from its mathematical objects to its sparse rank-wise message-passing code. It explains how combinatorial cells, typed neighborhood relations, geometric invariants, and equivariant coordinate updates are represented, and why the implementation exposes an explicit coordinate policy for data with no coordinates, graph-derived structural coordinates, or physical Euclidean coordinates. The report distinguishes the ETNN architecture introduced by Battiloro et al., behavior observed in the official NSAPH implementation, and framework-specific adaptations introduced by the TopoBench port. It is an implementation report, not a new formulation of ETNN. The implementation was developed for Track 2 of the 2026 Topological Deep Learning Challenge and validated on GraphUniverse and QM9 through complementary benchmark, integration, parity, and protocol-matched training studies.

Table of contents

Status and scope	3
1 Why ETNN requires more than ordinary graph message passing	3
1.1 A running picture: atoms, bonds, and higher-order cells	4
1.2 What E(n)-equivariance means here	4
2 From TopoBench batches to rank-wise ETNN state	5
2.1 The batch contract	5
2.2 Why placeholder compaction matters	6
3 Typed topological routes	6
4 Shared ETNN feature message passing	7
4.1 Relation-specific messages	7
4.2 Rank-wise residual updates	8
5 Why coordinates are an explicit policy	8
6 Policy 1: no coordinates	9
7 Policy 2: structural Laplacian coordinates	9
7.1 Rank-0 pseudo-coordinates	9
7.2 Lifting coordinates to higher-rank cells	10
7.3 Structural distance channels	10
8 Policy 3: physical coordinates	11
8.1 Recovering cell membership	11

8.2	Centroids and diameters	11
8.3	Directed Hausdorff-style channels	12
9	Invariant normalization	13
10	Equivariant coordinate updates	13
10.1	Radial displacement	14
10.2	Why the update is $E(n)$ -equivariant	14
11	Geometry must be recomputed between layers	15
12	End-to-end execution path	15
12.1	Construction time	15
12.2	Forward time	16
13	Faithfulness and implementation boundaries	16
14	Engineering decisions and trade-offs	17
14.1	Fixed relation order	17
14.2	Shape-stable empty relations	17
14.3	Relation-specific normalization	17
14.4	Dense membership and cell-wise distances	17
14.5	No automatic coordinate detection	17
14.6	Internal coordinate state	17
15	Validation across coordinate regimes	18
15.1	GraphUniverse: no coordinates versus structural coordinates	18
15.2	Physical-coordinate integration on QM9	19
15.3	Adapter and model-core parity audit	20
15.4	Protocol-matched QM9 training	21
15.5	Validation limits	22
16	Reading the source	22
17	What this implementation contributes	23
18	Conclusion	23
	References	24
19	Appendix: reproducibility and attribution record	24
19.1	Primary implementation	24
19.2	Original architecture	25
19.3	Reference software	25
19.4	Validation revisions and evidence records	25

Status and scope

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The corresponding implementation is submitted to TopoBench in [PR #391](#). The immutable revision cited above is hosted in the author’s public TopoBench fork while upstream review remains open.

This implementation was developed for Track 2 of the **2026 Topological Deep Learning Challenge: Bridging the Gap**, which asked participants to integrate topological neural networks into a shared TopoBench evaluation pipeline ([Geometric Intelligence Lab 2026](#)). The challenge provided the setting and common benchmark; the coordinate-policy abstraction, physical-coordinate extension, and layered validation strategy described here are the implementation developed for that setting.

The original architecture, mathematical framework, and equivariance results are described in the ETNN paper ([Battiloro et al. 2025](#)). The official implementation maintained by the NSAPH project is used as the reference for software-level behavior not fully fixed by the paper ([Battiloro et al. 2024](#)). TopoBench supplies the modular data, lifting, model, and readout contracts into which the implementation is integrated ([Telyatnikov et al. 2025](#)).

Short excerpts from the TopoBench implementation are reproduced to connect the mathematics to executable code. The repository is distributed under the MIT License; copyright remains with the TopoBench Team. Excerpts are lightly elided only where surrounding validation or documentation would obscure the specific mechanism under discussion.

Paper-derived formulas carry point-of-use citations with original locators; uncited displays are TopoBench-specific definitions or derivations.

1 Why ETNN requires more than ordinary graph message passing

Graph neural networks normally associate a feature vector with every vertex and exchange messages along edges. This representation is natural for pairwise interactions, but it does not give higher-order entities an independent state. A combinatorial complex can represent vertices, edges, faces, rings, functional groups, hyperedges, and other multi-way objects as **cells** with an integer rank ([Hajij et al. 2022](#)). ETNN performs message passing over those cells while incorporating geometric information from rank-0 Euclidean coordinates ([Battiloro et al. 2025](#)).

Using the combinatorial-complex formalism adopted by ETNN ([Hajij et al. 2022](#); [Battiloro et al. 2025, sec. 2](#)), let

$$\mathcal{C} = \bigsqcup_{r=0}^R \mathcal{C}_r$$

be a combinatorial complex whose rank- r cells form the set $\mathcal{C}_r$. The symbol R denotes the largest represented rank, and $n_r = |\mathcal{C}_r|$ denotes the number of rank- r cells. Every cell $c \in \mathcal{C}_r$ carries a feature vector $x_c \in \mathbb{R}^{d_r}$. In batched tensor form,

$$X_r \in \mathbb{R}^{n_r \times d_r}.$$

ETNN has two coupled responsibilities:

1. update features on cells of every visible rank through typed topological relations; and
2. when physical coordinates exist, derive invariant geometric channels and optionally update rank-0 positions equivariantly.

The first responsibility resembles heterogeneous message passing. The second is what makes a physical ETNN an $E(n)$ -equivariant geometric model rather than only a topological neural network.

1.1 A running picture: atoms, bonds, and higher-order cells

It is useful to keep one concrete complex in mind. Consider three atoms joined in a triangle. An ordinary graph stores three vertices and three pairwise edges. A combinatorial complex can treat the same objects as cells at different ranks:

- the atoms are rank-0 cells;
- the bonds are rank-1 cells; and
- the triangular ring, or another chosen molecular region, is a rank-2 cell.

The rank does not mean that one object is more important than another. It says how the objects are organized: a bond contains atoms, and a face or region can contain bonds. Each cell can carry its own state. Atom features might describe chemical species, bond features might describe bond type, and a higher-rank feature might summarize a ring or multi-atom interaction.

This picture turns ETNN implementation into three recurring questions:

1. **Who can communicate?** Topological relations decide whether information moves atom-to-atom, atom-to-bond, bond-to-ring, or in the reverse direction.
2. **What context accompanies a message?** Depending on the coordinate policy, an interaction may carry only a topological value, a structural distance, or physical summaries of the participating cells.
3. **What is allowed to move?** Scalar features are updated in every policy; physical rank-0 coordinates may also be updated, but only through an $E(n)$ -equivariant rule.

The rest of the report answers these questions in that order. The tensor and sparse-matrix details are implementations of this conceptual picture, not separate machinery.

1.2 What $E(n)$ -equivariance means here

The Euclidean group $E(n)$ contains translations, rotations, and reflections in n -dimensional space. Following the group action used in the original ETNN formulation (Battiloro et al. 2025, eq. 5), a transformation acts on a rank-0 position $p_i \in \mathbb{R}^n$ as

$$p_i \mapsto Qp_i + t, \quad Q^T Q = I,$$

where Q is orthogonal and t is a translation vector. Scalar cell features should remain invariant, while an updated coordinate should transform in the same way as the input coordinate. Relative vectors transform as

$$(p_i - p_j) \mapsto Q(p_i - p_j),$$

and Euclidean distances remain unchanged. ETNN uses invariant scalars to control vector-valued coordinate displacements, following the same basic equivariant construction used by $E(n)$ -equivariant graph networks (Satorras et al. 2021).

2 From TopoBench batches to rank-wise ETNN state

The mathematical complex must first become a collection of tensors that a neural network can process. TopoBench performs the graph-to-complex lifting and delivers cell features and sparse relations. ETNN's first task is therefore not to discover the complex, but to translate this framework representation into rank-indexed states and directed message edges without losing cell identity.

2.1 The batch contract

After a graph or another input domain is lifted to a combinatorial complex, TopoBench exposes rank-wise tensors:

```
1 x_0, x_1, ..., x_R
2 incidence_1, incidence_2, ..., incidence_R
3 up_adjacency_0, up_adjacency_1, ...
4 up_incidence_0, down_incidence_1, ...
```

The feature tensors contain one row per real cell. Sparse relation matrices use receiver cells on rows and sender cells on columns. ETNN instead consumes edge indices in [sender, receiver] order. Converting the representation correctly is therefore part of the model contract, not incidental tensor plumbing.

For relation N , let

$$A_N \in \mathbb{R}^{n_c \times n_d}$$

encode messages from sender rank d to receiver rank c . A nonzero entry $(A_N)_{ji}$ creates the directed message edge $i \rightarrow j$. The implementation coalesces the sparse tensor, removes explicitly stored zeros, converts matrix coordinates to sender/receiver indices, and preserves the scalar relation value where the selected coordinate policy requires it.

```
1 def _neighborhood_to_edge_index(
2     batch,
3     neighborhood: str,
4     src_rank: int,
5     dst_rank: int,
6     device: torch.device,
7     dtype: torch.dtype,
8     num_src_cells: int | None = None,
9     num_dst_cells: int | None = None,
10 ) → tuple[torch.Tensor, torch.Tensor]:
11     sparse_neighborhood = getattr(batch, neighborhood).coalesce()
12     indices = sparse_neighborhood.indices().long()
13     values = sparse_neighborhood.values()
14
15     nonzero_mask = values ≠ 0
16     indices = indices[:, nonzero_mask]
17     values = values[nonzero_mask]
18
19     receiver = dst_map[indices[0]]
20     sender = src_map[indices[1]]
21     valid_edge_mask = (receiver ≥ 0) & (sender ≥ 0)
22
```

```

23     edge_index = torch.stack(
24         [sender[valid_edge_mask], receiver[valid_edge_mask]], dim=0
25     ).to(device)
26     edge_attr = values[valid_edge_mask].view(-1, 1).to(
27         device=device, dtype=dtype
28     )
29     return edge_index, edge_attr

```

The complete implementation additionally checks for missing relation tensors, returns shape-stable empty tensors, and compacts placeholder slots introduced when a batch contains a graph with no cells at a requested rank.

2.2 Why placeholder compaction matters

Sparse batching and dense feature batching do not always have identical row counts. A graph with no rank-2 cells may receive one sparse placeholder slot so that block-diagonal relation matrices can be assembled, while x_2 contributes zero feature rows. Directly indexing x_2 with sparse indices would then be incorrect.

The helper `_sparse_axis_to_feature_index` reconstructs a mapping from sparse slots to real feature rows using `batch_r`. Real cells receive compact indices; placeholder slots receive -1; and edges touching placeholders are discarded. This explicit proof of alignment is preferable to silently clipping indices or assuming that every graph contains every visible rank.

3 Typed topological routes

The word *edge* can become ambiguous once the model contains cells of several ranks. A graph edge is a rank-1 object, while a message edge is simply a directed communication link between two cells. These are not the same thing. For example, an atom may send a message to an incident bond, two bonds may communicate because they bound the same face, and that face may send information back to each bond.

ETNN therefore does not collapse every communication pattern into one homogeneous relation. A neighborhood name identifies both the topological rule and the sender/receiver rank pair. For example:

Neighborhood	Sender rank	Receiver rank	Interpretation
up_adjacency-0	0	0	vertex-to-vertex adjacency
up_adjacency-1	1	1	edge-to-edge upper adjacency
up_incidence-0	0	1	vertex-to-edge incidence
down_incidence-1	1	0	edge-to-vertex incidence
up_incidence-1	1	2	edge-to-face incidence
down_incidence-2	2	1	face-to-edge incidence

TopoBench parses these names into ordered routes. The implementation preserves the configuration order because the destination update MLP observes incoming messages through concatenation. Changing route order after initialization would change the semantic meaning of input channels without changing their shape.

For every relation type N , ETNN creates a distinct message module ψ_N . For every destination rank r , it creates a rank-specific update module β_r . This separation lets two relations connecting the same pair of ranks learn different transformations.

4 Shared ETNN feature message passing

Once a route determines who communicates, the message layer decides what one cell should tell another. The same high-level recipe is used for every route: combine the sender state, receiver state, and scalar relation attributes; transform that combined description; learn how strongly the interaction should count; and sum the resulting messages at the receiver. Separate parameters let atom-to-bond communication learn a different function from bond-to-face communication even though both use this recipe.

4.1 Relation-specific messages

The following relation-resolved form implements the feature update in the original ETNN formulation (Battiloro et al. 2025, eq. 6). Let $h_d^{(\ell)} \in \mathbb{R}^h$ and $h_c^{(\ell)} \in \mathbb{R}^h$ denote sender and receiver embeddings at layer ℓ . Let $e_{d,c,N}^{(\ell)} \in \mathbb{R}^{q_N}$ be the scalar edge attributes supplied by the coordinate policy. The ungated edge message is

$$\tilde{m}_{d,c,N}^{(\ell)} = \phi_N^{(\ell)} \left(\left[h_d^{(\ell)} \parallel h_c^{(\ell)} \parallel e_{d,c,N}^{(\ell)} \right] \right),$$

where $\parallel$ denotes concatenation and $\phi_N^{(\ell)} : \mathbb{R}^{2h+q_N} \rightarrow \mathbb{R}^h$ is a relation-specific MLP.

Not every topologically permitted interaction is equally informative. The ETNN paper specifies adjacency-wise weighted aggregation, with nonnegative scalar weights produced from each message (Battiloro et al. 2025, app. K). The official NSAPH implementation realizes that mechanism as a learned scalar edge gate. The gate behaves like a soft relevance score between zero and one: it can suppress a message without removing the underlying relation from the complex. The TopoBench port retains that behavior:

$$g_{d,c,N}^{(\ell)} = \sigma \left(W_{g,N}^{(\ell)} \tilde{m}_{d,c,N}^{(\ell)} + b_{g,N}^{(\ell)} \right),$$

$$m_{c,N}^{(\ell)} = \sum_{d \in N(c)} g_{d,c,N}^{(\ell)} \tilde{m}_{d,c,N}^{(\ell)}.$$

Here $W_{g,N}^{(\ell)} \in \mathbb{R}^{1 \times h}$, $b_{g,N}^{(\ell)} \in \mathbb{R}$, σ is the logistic sigmoid, and $N(c)$ denotes the senders connected to receiver c by relation N .

```

1 sender, receiver = edge_index
2 state = torch.cat(
3     [x_src[sender], x_dst[receiver], edge_attr.to(x_dst.dtype)],
4     dim=-1,
5 )
6 messages = self.message_mlp(state)
7 messages = messages * self.edge_gate(messages)
8 out.index_add_(0, receiver, messages)

```

`index_add_` implements sum aggregation without materializing a dense relation matrix. Empty relations return an all-zero tensor with one row per destination cell, preserving the downstream update shape.

4.2 Rank-wise residual updates

Suppose rank r receives the ordered relations $N_1, \dots, N_{k_r}$. The residual update below is the TopoBench tensor form of the rank- and neighborhood-dependent ETNN update (Battiloro et al. 2025, eq. 6 and app. K):

$$h_c^{(\ell+1)} = h_c^{(\ell)} + \beta_r^{(\ell)} \left(h_c^{(\ell)} \parallel m_{c,N_1}^{(\ell)} \parallel \dots \parallel m_{c,N_{k_r}}^{(\ell)} \right).$$

Because incoming relation counts are known at construction time, the input width of β_r is $(1+k_r)h$. The implementation builds an update MLP for every visible rank, including a rank with no incoming configured relation. Such a rank still receives a learned residual transformation of its current state rather than causing a missing-module failure.

```

1 incoming_counts = defaultdict(int)
2 for _, dst_rank in self.routes:
3     incoming_counts[dst_rank] += 1
4
5 self.update = nn.ModuleDict(
6     {
7         str(rank): _make_mlp(
8             in_channels=(1 + incoming_counts[rank]) * hidden_channels,
9             hidden_channels=hidden_channels,
10            out_channels=hidden_channels,
11            dropout=dropout,
12            activation=activation,
13            use_batch_norm=use_batch_norm,
14        )
15     for rank in range(max_rank + 1)
16 }
17 )

```

5 Why coordinates are an explicit policy

An implementation that simply checks whether a batch happens to contain a `pos` attribute can silently change model semantics across datasets. A field named `pos` may contain a physical configuration, a visualization layout, or a precomputed embedding. Conversely, a coordinate-free dataset should not be assigned fabricated physical geometry merely to satisfy an API.

The backbone therefore requires one explicit policy:

Policy	Coordinate source	Edge attributes	Coordinate dynamics
none	none	sparse relation value	none
structural_laplace	graph Laplacian eigenvectors	relation value and squared structural distance	fixed
physical	rank-0 Euclidean positions	physical cell invariants	optional equivariant update

This is an implementation extension, not a claim that all three modes realize the same physical symmetry. Only the physical policy uses genuine Euclidean coordinates and implements coordinate dynamics. The structural policy

is invariant to orthogonal changes of basis in its chosen pseudo-coordinate frame, but that does not endow the original graph with a physical $E(n)$ action.

The policy is validated during construction. Configuration errors fail early: `pos_update=True` is rejected outside physical mode, physical invariant normalization is rejected for nonphysical policies, and the coordinate-update relation must be a configured rank-0-to-rank-0 route.

6 Policy 1: no coordinates

The coordinate-free mode isolates ETNN’s typed topological feature update. For each nonzero sparse relation entry,

$$e_{d,c,N} = [a_{d,c,N}],$$

where $a_{d,c,N}$ is the stored relation value. No Euclidean claim is made, and no coordinate tensor is required.

This mode is useful when the data expresses topology but no meaningful physical frame. It is intentionally conservative: absence of geometry changes the model’s information set rather than triggering an undocumented surrogate.

```
1 if self.coordinate_policy == "none":
2     return edge_attr
```

7 Policy 2: structural Laplacian coordinates

Some datasets have no physical coordinates but still contain meaningful notions of graph position. A vertex near the center of a community, for example, plays a different structural role from a vertex near its boundary even if neither has a location in Euclidean space. Laplacian positional encodings turn this global connectivity information into numerical pseudo-coordinates. They let the message function compare structural position while keeping the distinction between graph geometry and physical geometry explicit.

7.1 Rank-0 pseudo-coordinates

The graph Laplacian can be viewed as an operator that measures how a signal varies across adjacent vertices. Its low-frequency eigenvectors are smooth signals over the graph: nearby or similarly situated vertices tend to receive similar values. Selecting several eigenvectors assigns every vertex a short vector describing its position within these global structural patterns (Dwivedi et al. 2023). Let

$$P_0 \in \mathbb{R}^{n_0 \times k}$$

contain k selected eigenvectors, with row p_v assigned to rank-0 cell v . These values encode graph structure, not physical position.

The transform stores the matrix separately from input features. Setting `concat_to_x=false` is important: concatenating the eigenvectors into `x_0` would change the feature encoder and prevent the backbone from controlling how structural coordinates enter messages.

7.2 Lifting coordinates to higher-rank cells

Only vertices receive Laplacian coordinates directly. A bond, face, or other higher-rank cell still needs a structural location if it is to participate in distance-aware messages. The natural construction is a barycenter: a bond is placed at the average of its endpoint coordinates, and a face is placed at the average of its boundary-cell coordinates. Recursively, for rank $r > 0$, the implementation averages the coordinates of incident rank- $(r - 1)$ cells. Let $B_r \in \mathbb{R}^{n_{r-1} \times n_r}$ be the incidence matrix between ranks $r - 1$ and r . For a rank- r cell c ,

$$p_c^{(r)} = \frac{\sum_{d=1}^{n_{r-1}} |(B_r)_{dc}| p_d^{(r-1)}}{\sum_{d=1}^{n_{r-1}} |(B_r)_{dc}|}.$$

Absolute incidence values are used because orientation signs should not cancel coordinates during barycentric averaging.

```

1 indices = incidence.indices()
2 values = incidence.values().abs().to(lower_coordinates.dtype)
3 lower_idx, cell_idx = indices
4
5 coordinates.index_add_(
6     0,
7     cell_idx,
8     lower_coordinates[lower_idx] * values.unsqueeze(-1),
9 )
10 weights.index_add_(0, cell_idx, values.unsqueeze(-1))
11 return coordinates / weights.clamp_min(torch.finfo(weights.dtype).eps)

```

7.3 Structural distance channels

These pseudo-coordinates are never shown to the model component-by-component. Doing so would make predictions depend on arbitrary eigenvector signs or basis choices. Instead, the model receives their squared separation, which answers a more stable question: *how far apart are these two cells in the selected structural embedding?* For a relation edge from cell d to cell c , the policy supplies

$$e_{d,c,N} = \left[\frac{a_{d,c,N}}{\|p_d - p_c\|_2^2} \right].$$

The squared distance is invariant under a consistent translation and orthogonal transformation of the structural frame. It is consequently insensitive to global eigenvector sign flips and to orthogonal basis changes inside a degenerate eigenspace, provided the frame is transformed consistently for the entire complex.

```

1 sender, receiver = edge_index
2 delta = src_coordinates[sender] - dst_coordinates[receiver]
3 distance_attr = delta.pow(2).sum(dim=-1, keepdim=True)
4 return torch.cat([edge_attr, distance_attr], dim=-1)

```

The policy deliberately keeps these coordinates fixed. Updating Laplacian eigenvectors as though they were physical positions would destroy their meaning as a deterministic structural encoding.

8 Policy 3: physical coordinates

Physical mode begins from a different premise: rank-0 coordinates are measured positions in Euclidean space, such as atomic coordinates in a molecule. A higher-rank cell does not usually arrive with its own position. It represents a set of vertices, so its geometry must be derived from that set. The implementation first reconstructs which vertices belong to each cell and then summarizes the resulting point set in ways that are unchanged by translation, rotation, or reflection.

8.1 Recovering cell membership

The physical policy starts from rank-0 positions

$$P \in \mathbb{R}^{n_0 \times n},$$

where row p_i is the physical coordinate of vertex i . Higher-rank cell geometry depends on which rank-0 vertices belong to each cell. TopoBench provides consecutive incidence matrices rather than a direct vertex-to-cell membership matrix at every rank, so the implementation reconstructs

$$M_r \in \{0, 1\}^{n_0 \times n_r}.$$

The base case is $M_0 = I_{n_0}$. Recursively,

$$M_r = \mathbf{1}(M_{r-1}|B_r| > 0),$$

where the indicator is applied elementwise. This computes transitive rank-0 membership while ignoring incidence orientation signs.

Dense membership is simple and robust for small and medium physical complexes, but it scales as $O(n_0 n_r)$ per rank. That trade-off is documented rather than hidden: a large-complex implementation should replace it with sparse membership propagation.

8.2 Centroids and diameters

Two coarse summaries establish where a cell lies and how large it is. The centroid is the average position of its vertices; the diameter is the largest distance between any two of them. Thus an atom is a zero-diameter singleton, a bond has its midpoint as centroid and bond length as diameter, and a larger cell spans the vertices on its boundary. Neither summary depends on the orientation of the global coordinate frame. The centroid construction specializes the permutation-invariant geometric summary used in the ETNN paper ([Battiloro et al. 2025](#), eq. 10).

For a cell c , let

$$V(c) = \{i : (M_r)_{ic} = 1\}$$

be its incident rank-0 vertices. Its centroid is

$$\mu_c = \frac{1}{|V(c)|} \sum_{i \in V(c)} p_i,$$

and its diameter is

$$\delta_c = \max_{i,j \in V(c)} \|p_i - p_j\|_2.$$

Singleton cells have zero diameter. Empty malformed cells retain zero-valued summaries so tensor shapes remain valid, although valid lifted complexes should not contain such cells.

The centroid computation is vectorized as

$$\mu_r = M_r^T P \oslash \max(M_r^T \mathbf{1}, 1),$$

where $\oslash$ denotes row-wise division. Diameters are computed cell by cell with `torch.cdist`, reflecting the typically small cardinality of molecular or geometric cells.

8.3 Directed Hausdorff-style channels

Centroids and diameters do not fully describe how two cells occupy space. Two point sets can share a centroid and diameter while arranging their vertices differently. A directed Hausdorff quantity asks a complementary coverage question: *how far must one look, in the worst case, to match every vertex of the sender cell to some vertex of the receiver cell?*

The computation has two nested steps. For each sender vertex, find the nearest receiver vertex. Then take the largest of those nearest-neighbor distances. A small result means every sender vertex lies close to the receiver set; a large result means at least one sender vertex is poorly covered. The report calls these channels *Hausdorff-style* because they operate on the finite rank-0 vertex sets representing cells, rather than on every point of a continuous geometric region.

For sender cell d and receiver cell c , define the directed term appearing inside the paper’s symmetric Hausdorff invariant (Battiloro et al. 2025, eq. 11):

$$H(d, c) = \max_{i \in V(d)} \min_{j \in V(c)} \|p_i - p_j\|_2.$$

The direction matters. If the vertices of d are contained among the vertices of c , then $H(d, c) = 0$: every vertex of d has an exact match in c . However, $H(c, d)$ can be positive because c may contain additional vertices far from d . The conventional symmetric Hausdorff distance would retain only the larger of the two values. ETNN instead keeps both directed quantities, so the message function can distinguish containment-like relationships without using coordinate components directly. In general, $H(d, c) \neq H(c, d)$.

The ETNN paper identifies centroid distance and the two directed Hausdorff terms as valid geometric invariants (Battiloro et al. 2025, eqs. 10–11). The particular three- or five-channel vector used here, including the sender and receiver diameters, follows the official reference implementation (Battiloro et al. 2024):

$$e_{d,c,N} = \begin{bmatrix} \|\mu_d - \mu_c\|_2 \\ \delta_d \\ \delta_c \\ H(d, c) \\ H(c, d) \end{bmatrix}.$$

Without the final two channels, it has width three. Unlike the other policies, physical mode does **not** append the sparse TopoBench relation value. The sparse operator determines which pairs communicate; the edge attributes follow the geometric invariant construction used by the NSAPH implementation.

```

1 invariant_attr = _physical_relation_invariants(
2     src_centroids=physical_geometry.centroids[src_rank],
3     dst_centroids=physical_geometry.centroids[dst_rank],
4     src_diameters=physical_geometry.diameters[src_rank],
5     dst_diameters=physical_geometry.diameters[dst_rank],
6     src_membership=physical_geometry.vertex_memberships[src_rank],
7     dst_membership=physical_geometry.vertex_memberships[dst_rank],
8     vertex_coordinates=physical_geometry.vertex_coordinates,
9     edge_index=edge_index,
10    dtype=edge_attr.dtype,
11    hausdorff_dists=self.hausdorff_dists,
12 )

```

9 Invariant normalization

The geometric channels answer different questions and need not share a numerical scale. A centroid separation can be much larger than a cell diameter, and an atom-to-atom relation can have a different distribution from a bond-to-face relation. If these raw values enter an MLP unchanged, the largest scale can dominate optimization simply because of its units. Normalization puts the channels on comparable working scales while preserving their role as invariant scalars. Because each topological relation has its own distribution, normalization is relation-specific.

The most faithful option uses one non-affine BatchNorm1d module per relation, following the paper’s architecture description and the official implementation (Battiloro et al. 2025, app. K; Battiloro et al. 2024):

```

1 self.invariant_batch_norm = nn.ModuleList(
2     [
3         nn.BatchNorm1d(self.edge_channels, affine=False)
4         for _ in self.neighborhoods
5     ]
6 )

```

An empty relation bypasses normalization. A singleton relation bypasses training-mode BatchNorm because a one-row batch cannot estimate variance; in evaluation mode it can still use accumulated running statistics.

The implementation also offers detached mean-absolute normalization:

$$\hat{e}_{:,k} = \frac{e_{:,k}}{\max(\text{mean}|e_{:,k}|, \epsilon)}.$$

The denominator is detached so normalization rescales the forward values without introducing gradients through the batch statistic. This is a framework-safe alternative, not the default reference behavior.

10 Equivariant coordinate updates

Feature updates change what the model knows about a vertex. A physical ETNN can also change where that vertex is represented internally, allowing later layers to reason about a learned geometry. The constraint is stronger than for an ordinary coordinate regressor: rotating, reflecting, or translating the input must rotate, reflect, or translate the updated coordinates in exactly the same way.

ETNN satisfies this by separating *magnitude* from *direction*. Invariant feature messages determine a scalar amount of movement. Relative coordinate vectors supply directions that transform correctly under $E(n)$. Multiplying the two cannot introduce a preferred global axis.

10.1 Radial displacement

When `pos_update=True`, one configured rank-0-to-rank-0 relation drives an internal coordinate update. This is the implementation’s relation-specific realization of the ETNN coordinate rule (Battiloro et al. 2025, eq. 7). Let $m_i^{(\ell)} \in \mathbb{R}^h$ be the aggregated rank-0 message associated with vertex i . A bias-free linear map produces a scalar radial weight:

$$w_i^{(\ell)} = W_x^{(\ell)} m_i^{(\ell)}, \quad W_x^{(\ell)} \in \mathbb{R}^{1 \times h}.$$

For a message edge $i \rightarrow j$, the implementation applies the receiver-side weight along the sender-receiver relative vector and scatters the displacement to the sender:

$$\Delta p_i^{(\ell)} = \sum_{j:(i,j) \in E_x} (p_i^{(\ell)} - p_j^{(\ell)}) w_j^{(\ell)},$$

$$p_i^{(\ell+1)} = p_i^{(\ell)} + \alpha \Delta p_i^{(\ell)},$$

where E_x is the selected coordinate-update relation and α is a fixed scale.

```

1 radial_weight = self.coordinate_update_mlp(
2     aggregated_message[receiver]
3 ).to(coordinates.dtype)
4 relative = coordinates[sender] - coordinates[receiver]
5 coordinate_delta = coordinates.new_zeros(coordinates.shape)
6 coordinate_delta.index_add_(0, sender, relative * radial_weight)
7 return coordinates + self.coordinate_update_scale * coordinate_delta

```

The map is a single bias-free linear layer initialized with a truncated normal distribution, matching the reference implementation more closely than a deeper MLP would.

10.2 Why the update is $E(n)$ -equivariant

The message and radial weight depend only on invariant scalar features. The argument below restates the equivariance result established for ETNN layers (Battiloro et al. 2025, theorem 1 and app. E). Under $p_i \mapsto Qp_i + t$,

$$(p_i - p_j)w_j \mapsto Q(p_i - p_j)w_j.$$

Therefore

$$\Delta p_i \mapsto Q\Delta p_i$$

and

$$p_i + \alpha \Delta p_i \mapsto Q(p_i + \alpha \Delta p_i) + t.$$

The feature path remains invariant and the coordinate path remains equivariant.

11 Geometry must be recomputed between layers

Coordinate updates change more than the vertex positions themselves. Moving an atom changes the midpoint and length of its bonds, the centroid and diameter of larger cells, and the directed Hausdorff relationships between cells. If those summaries remain cached, a later layer would reason about the old geometry while using the new coordinates. The physical forward pass therefore follows this order:

1. validate and copy rank-0 physical positions;
2. reconstruct cell memberships;
3. compute geometry from the current positions;
4. perform one ETNN feature/message layer;
5. update rank-0 positions when enabled;
6. recompute all physical invariants before the next layer.

Membership is topological and remains fixed. Geometry is coordinate-dependent and is rebuilt.

```
1 physical_geometry = _build_physical_cell_geometry(  
2     batch=batch,  
3     vertex_coordinates=physical_coordinates,  
4     max_rank=self.max_rank,  
5 )  
6  
7 for layer in self.layers:  
8     x, physical_coordinates = layer(  
9         x=x,  
10        batch=batch,  
11        structural_coordinates=None,  
12        physical_geometry=physical_geometry,  
13        physical_coordinates=physical_coordinates,  
14    )  
15     if self.pos_update:  
16         physical_geometry = _summarize_physical_cell_geometry(  
17             vertex_coordinates=physical_coordinates,  
18             vertex_memberships=physical_geometry.vertex_memberships,  
19         )
```

The updated coordinates remain internal to the backbone. The TopoBench wrapper and readout consume rank-wise feature embeddings, so the model does not mutate the input batch or silently change the public pipeline contract.

12 End-to-end execution path

The complete forward pass can be summarized as follows.

12.1 Construction time

1. Parse configured neighborhood names into sender/receiver routes.
2. Validate the explicit coordinate policy and normalization mode.
3. Determine policy-dependent edge width: one, two, three, or five channels.
4. Construct one message module per relation and one update module per rank.
5. For physical coordinate dynamics, construct the bias-free radial map.

12.2 Forward time

1. Collect visible x_r tensors and project them into the shared hidden width.
2. Build fixed structural coordinates or initial physical geometry if required.
3. For each ETNN layer and each configured relation:
 - a. convert the sparse relation into compact message edges;
 - b. construct policy-specific scalar edge attributes;
 - c. compute gated relation messages;
 - d. aggregate messages into the destination rank;
 - e. update physical coordinates on the selected rank-0 relation.
4. Concatenate incoming messages rank-wise and apply residual feature updates.
5. Recompute physical geometry after each coordinate-changing layer.
6. Apply rank-wise output projections and return `dict[int, Tensor]`.

The wrapper subsequently exposes those tensors through TopoBench’s expected x_0 , x_1 , and x_2 fields. A signal-down readout can then propagate higher-rank representations toward rank 0 before task-specific pooling. These components surround the backbone but do not alter ETNN’s internal coordinate semantics.

13 Faithfulness and implementation boundaries

The word *faithful* needs a precise object. The implementation can be faithful to a paper equation, a reference repository, or a host framework contract; those targets do not always determine every detail identically.

Mechanism	ETNN paper	NSAPH software	TopoBench implementation
rank-wise topological messages	defining architecture	concrete modules	relation-specific sparse modules
scalar edge gate	implementation-level realization	linear + sigmoid	retained
physical invariants	centroid and cell geometry	3/5 channel construction	retained in physical mode
coordinate update	equivariant radial update	bias-free linear scalar map	retained
invariant refresh	required when positions move	recomputed between layers	retained
no-coordinate behavior	not the physical ETNN setting	not the primary path	explicit conservative policy
Laplacian coordinates	not physical ETNN	not the primary path	explicit structural extension
sparse batching	implementation-independent	native data representation	TopoBench relation adapter
placeholder ranks	framework-independent	not a TopoBench concern	validated and compacted

This table separates adaptation from invention. The coordinate-free and structural policies reuse ETNN’s rank-wise feature-message scaffold but do not claim to reproduce physical ETNN. The physical policy is the implementation path aligned most closely with the original architecture.

14 Engineering decisions and trade-offs

14.1 Fixed relation order

Incoming relation messages are concatenated, so configuration order is part of the learned function. The code treats routes as immutable tuples and validates one route per neighborhood.

14.2 Shape-stable empty relations

An empty relation still contributes a zero tensor of shape $n_{\text{dst}} \times h$. This keeps update dimensions fixed across batches and avoids data-dependent module structure.

14.3 Relation-specific normalization

Physical invariants from vertex adjacency and edge-face incidence can have different distributions. Independent normalizers prevent one relation from overwriting another relation's running statistics.

14.4 Dense membership and cell-wise distances

Dense vertex-to-cell membership and Python-level loops over cells or relation edges prioritize correctness and clarity for small physical complexes. Their cost should be made explicit:

- membership storage grows as $O(n_0 n_r)$;
- diameter computation performs pairwise distances inside every cell;
- directed Hausdorff computation performs two set-to-set reductions per relation edge;
- converting device indices to Python lists can synchronize accelerators.

These are implementation limits, not equivariance limits. A future scalable version could retain identical mathematics while using segmented sparse reductions and batched ragged distance kernels.

14.5 No automatic coordinate detection

An auto mode would be convenient, but convenience is outweighed by semantic ambiguity. Explicit policies make configurations reproducible and ensure that a layout coordinate is not silently treated as physical geometry.

14.6 Internal coordinate state

The backbone copies input positions before updating them and does not overwrite `batch.pos`. This prevents surprising interactions with other transforms, wrappers, or repeated forward calls.

15 Validation across coordinate regimes

The coordinate-policy design creates a validation problem that a single dataset cannot resolve. GraphUniverse supplies controlled graph families and a common challenge benchmark, but its graphs do not carry physical Euclidean coordinates (Van Langendonck et al. 2026). It can therefore evaluate the none and structural_lappe policies, but it cannot exercise the physical geometry contract. QM9 supplies atomic positions and molecular targets (Ramakrishnan et al. 2014), so it is used for the complementary physical-coordinate study.

The resulting validation is layered:

1. **GraphUniverse benchmark:** compare the coordinate-free and structural policies across controlled graph regimes.
2. **Physical-path integration:** execute the public TopoBench physical policy from molecular positions through lifting, ETNN, readout, and loss.
3. **Adapter audit:** verify that native QM9CC tensors are translated into the TopoBench model contract without changing their contents.
4. **Model-core parity:** compare native and TopoBench forward, backward, and optimizer-step behavior under shared inputs and initialization.
5. **Protocol-matched training:** compare the native reference and the TopoBench ETNN core parity harness over the paper-aligned 1,000-epoch QM9 protocol.

These checks answer different questions. Completion of the physical path does not establish predictive quality; a parity audit does not validate an independent lifting; and a single-seed training comparison does not establish general superiority.

15.1 GraphUniverse: no coordinates versus structural coordinates

GraphUniverse generates families of graphs with persistent semantic structure while controlling properties such as homophily and degree distributions (Van Langendonck et al. 2026). The challenge evaluation used two tasks: community detection, measured by accuracy, and triangle counting, measured by mean squared error normalized by the total number of triangles. Each policy was evaluated for 12 combinations of homophily, average-degree, and power-law regimes with seeds 42, 43, and 44. This gives 36 paired observations per task and 72 runs per policy.

Task	No coordinates	Structural coordinates	Mean structural advantage	Descriptive paired interval
Community detection accuracy	0.4535 ± 0.1299	0.4543 ± 0.1313	+0.00084	[-0.00064, 0.00229]
Triangle MSE / total triangles	0.1153 ± 0.1009	0.1211 ± 0.1042	-0.00578	[-0.02658, 0.01357]

For accuracy, structural advantage is structural minus coordinate-free performance. For triangle counting, the sign is reversed so that positive still means structural coordinates are better. The pooled mean and standard deviation summarize all settings and seeds. The final column gives the 2.5th and 97.5th percentiles of 20,000 bootstrap resamples of the 36 observed paired differences. Because the 12 graph regimes form a fixed factorial grid crossed with three seeds, these are descriptive intervals for the evaluated grid, not population-level confidence intervals. Both intervals include zero, so the study supports functional comparability rather than a universal structural coordinate advantage.

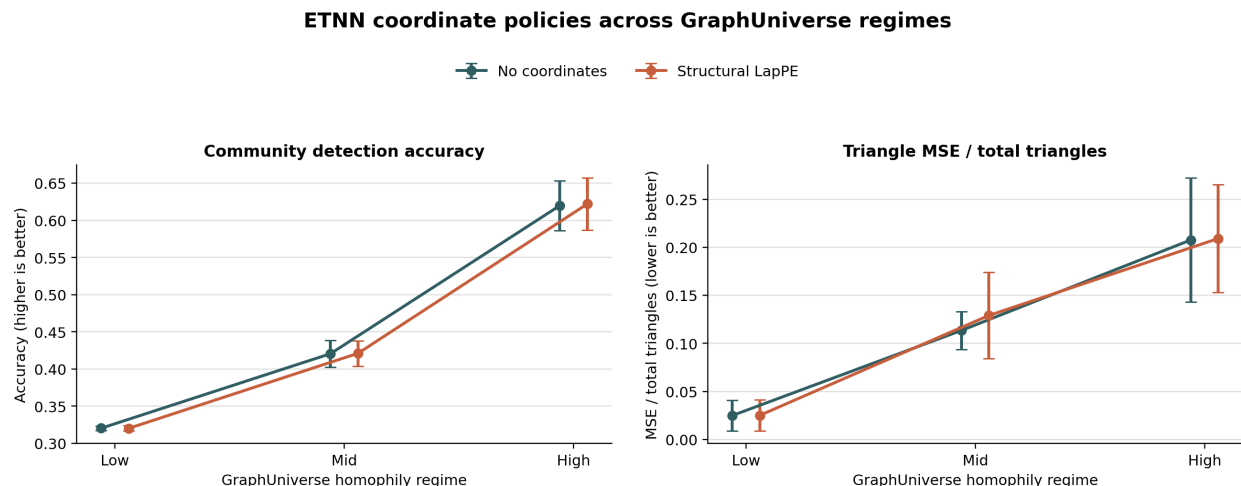


Figure 1: In-distribution GraphUniverse performance by homophily regime. Points average the four degree and power-law settings over three seeds; error bars show 1.96 standard errors across the 12 observed regime-seed values and are descriptive rather than population confidence intervals.

The homophily effect is much larger than the gap between coordinate policies. Both policies improve substantially on community detection as homophily rises, while normalized triangle-counting error also increases with the graph regime.

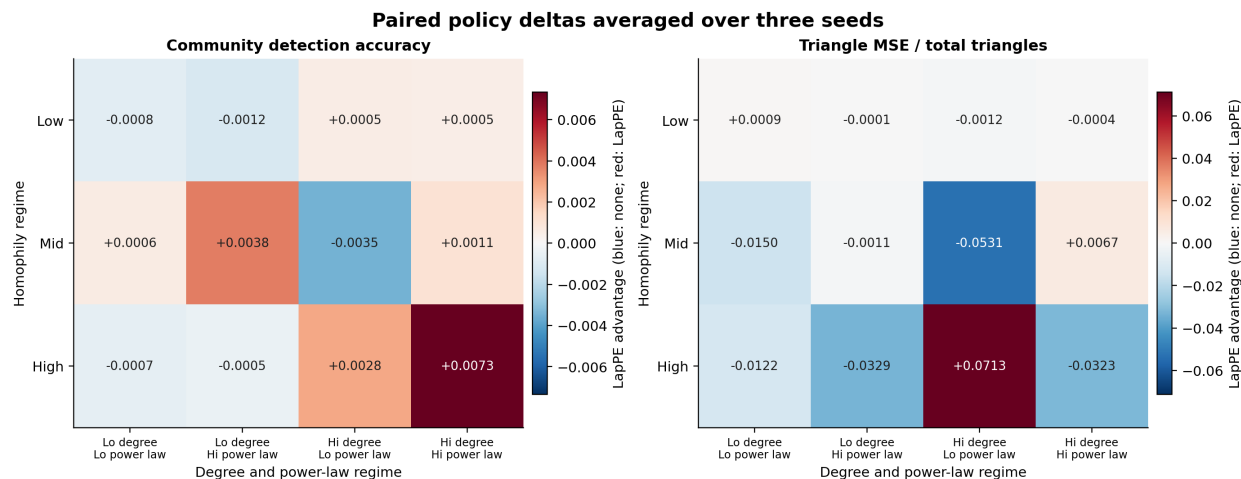


Figure 2: Paired structural-coordinate advantage across the 12 GraphUniverse regimes, averaged over three seeds. Red denotes an advantage for structural Laplacian coordinates; blue denotes an advantage for no coordinates.

The paired view reveals local gains and losses that disappear in a global average. Structural coordinates win 23 of 36 community-detection pairs and 16 of 36 triangle-counting pairs, but the direction depends on the regime. This is consistent with the small global effects and wide uncertainty intervals.

15.2 Physical-coordinate integration on QM9

QM9 contains equilibrium molecular geometries and quantum-chemical properties for approximately 134,000 small organic molecules (Ramakrishnan et al. 2014). The public TopoBench physical configuration reads rank-0

coordinates from pos, applies `GraphTriangleInducedCC`, constructs physical relation invariants, and enables both learned position updates and directed Hausdorff-style channels.

A reduced end-to-end integration study executed three physical variants to completion. Each used seed 42, batch size 128, 10 epochs, 50 training batches, 10 validation batches, and 10 test batches per epoch.

Variant	Position updates	Hausdorff channels	Test MAE (D)	End-to-end elapsed time
Fixed coordinates	no	yes	0.818540	58 min 28 s
Dynamic coordinates	yes	no	0.873744	11 min 44 s
Full physical policy	yes	yes	0.868965	1 h 39 min 21 s

The elapsed times are single-run wall-clock observations that include framework overhead; the first variant also includes first-use dataset preparation. They are not hardware-normalized benchmarks. Their large spread nevertheless records the practical cost of the per-relation Hausdorff construction in the public physical path. Peak memory was not separately profiled.

This study was deliberately bounded in epochs and batches. Its purpose was to verify the public `QM9` $\rightarrow$ `lifting` $\rightarrow$ `physical ETNN` $\rightarrow$ `readout` $\rightarrow$ `loss` path; its predictive metrics are therefore not used for scientific comparison.

The stronger quantitative study controls the outer protocol and isolates the ETNN model core. Native `QM9CC` atoms, bonds, whole-molecule cells, five-channel physical invariants, train/validation/test splits, target normalization, initial model state, optimizer, and learning-rate schedule are shared between the pinned `NSAPH` implementation and the reproduction-specific `ETNNQM9Parity` harness. This harness reuses `_ETNNMessagePassing` and `_make_mlp` from the public coordinate-policy implementation while matching the native feature embedding, normalization, pooling, and readout contract. This design asks whether the shared `TopoBench ETNN` core preserves reference behavior. It does not train the public `ETNNCoordinatePolicy` class end to end and, because it deliberately reuses native construction and invariants, it is not an independent benchmark of the `TopoBench` molecular lifting.

15.3 Adapter and model-core parity audit

Before long training, an eight-molecule native batch was translated into the `TopoBench` model contract. All 25 adapter checks passed: rank-wise features, positions, targets, batch assignments, relation indices, and invariant tensors matched the canonical native values and were independently stored.

The subsequent model-core audit loaded an identical 1,497,871-parameter state and compared evaluation mode, training mode, loss, gradients, buffers, one Adam step, and post-step outputs.

Audit quantity	Maximum absolute difference	Result
Evaluation output	2.98e-8	pass
Training output	2.98e-8	pass
Loss	0	pass
Training buffers	0	pass
Gradients	3.73e-8	pass
Parameters after one Adam step	2.17e-6	strict check failed in 4 tensors
Output after one Adam step	2.98e-8	pass

Eight of nine model checks passed. The one failure was not hidden: four tensors exceeded the $1\text{e-}6$ optimizer tolerance after one update, with a worst absolute difference of $2.17\text{e-}6$. The post-step predictions nevertheless remained within $2.98\text{e-}8$. The audit therefore demonstrates near-identical functional behavior at the forward, backward, and output levels, but not bitwise-equivalent parameter updates.

15.4 Protocol-matched QM9 training

The final controlled comparison used the dipole-moment target μ , the EGNN split, seed 42, seven ETNN layers with 128 hidden channels, batch size 96, learning rate 5×10^{-4} , weight decay 10^{-5} , gradient clipping at 1.0, and 1,000 epochs. These settings reproduce the paper-aligned experiment 1 protocol rather than the shorter defaults of the pinned repository.

Implementation	Best validation MAE (D)	Selected epoch	Test MAE (D)
ETNN paper, Table 11, configuration 1	not reported	not reported	0.030
Native NSAPH reproduction	0.030985	1000	0.031815
TopoBench ETNN core parity harness (ETNNQM9Parity)	0.030401	999	0.031217

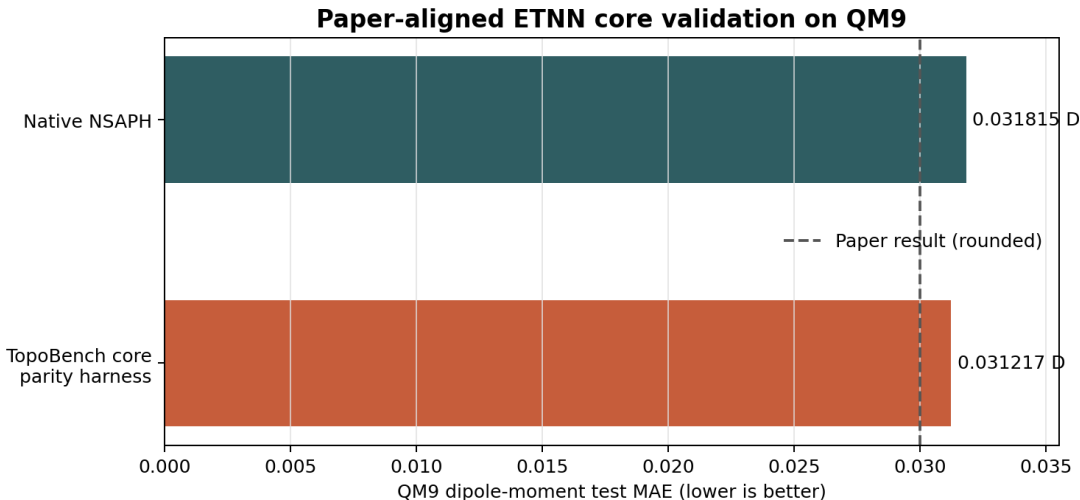


Figure 3: QM9 dipole-moment test MAE for the native NSAPH reproduction and the TopoBench ETNN core parity harness. The dashed line is the paper value, which is reported to three decimal places.

The TopoBench core parity harness differs from the native reproduction by -0.000598 D, or -1.88% , in test MAE. That difference is small relative to the precision of the rounded paper result and should not be interpreted as evidence that one implementation is generally superior. The controlled result supports training-level fidelity of the shared ETNN core under one seed and protocol.

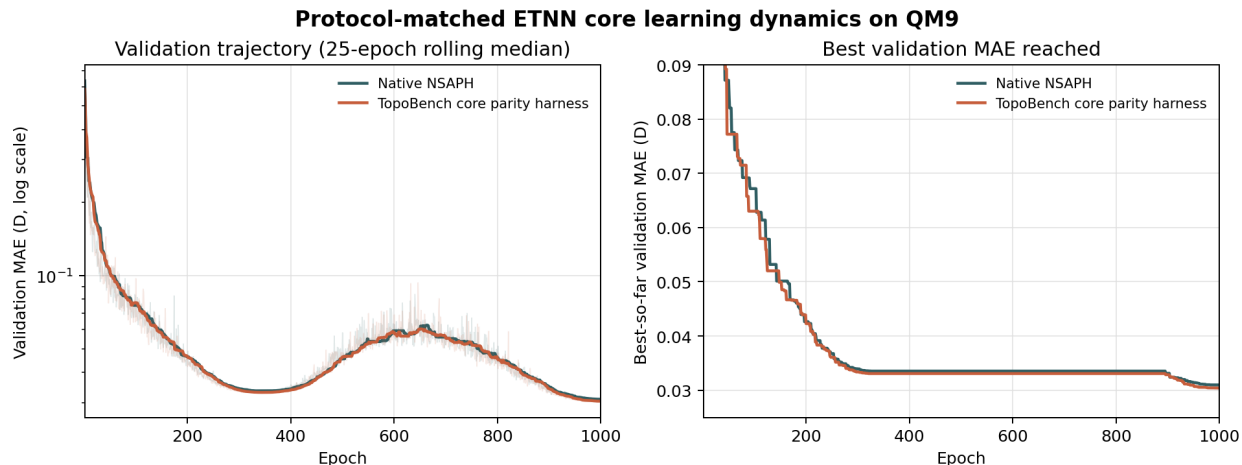


Figure 4: Native and TopoBench ETNN core parity-harness validation trajectories over the matched 1,000-epoch QM9 protocol. The left panel shows raw values faintly and a 25-epoch rolling median; the right panel shows best-so-far validation MAE.

The learning curves follow the same schedule-driven shape. Their 25-epoch rolling medians have correlation 0.9964 and mean absolute separation 0.00125 D. These are descriptive trajectory summaries, not independent statistical replicates. Taken together with the adapter and parity audits, the result supports a narrow conclusion: the TopoBench ETNN model core preserves the behavior of the reference implementation closely enough to reproduce its QM9 training scale under controlled inputs.

15.5 Validation limits

The validation evidence should be read with four boundaries in mind:

1. GraphUniverse evaluates the none and structural_lappe policies, not physical geometry.
2. The reduced public-path QM9 study checks integration but is not a paper-comparable performance experiment.
3. The protocol-matched study substitutes only the model core and reuses native QM9CC construction and invariants.
4. The 1,000-epoch comparison uses one seed, so it establishes reproducibility at that seed rather than a population-level performance claim.

16 Reading the source

The consolidated implementation is organized by responsibility rather than by policy-specific subclasses:

Source region	Responsibility
<code>_get_activation, _make_mlp</code>	shared neural blocks
<code>_ETNNMessagePassing</code>	gated relation message and aggregation
<code>_sparse_axis_to_feature_index</code>	placeholder-safe sparse compaction
<code>_neighborhood_to_edge_index</code>	TopoBench relation conversion
<code>_build_lappe_cell_coordinates</code>	structural coordinate lifting
<code>ETNNCoordinatePolicy</code>	policy selection and multi-layer orchestration
<code>_ETNNCoordinatePolicyLayer</code>	one feature/coordinate update layer

Source region	Responsibility
<code>_build_vertex_memberships</code>	rank-0 membership reconstruction
<code>_compute_cell_centroids</code>	invariant cell centers
<code>_compute_cell_diameters</code>	invariant cell extents
<code>_physical_relation_invariants</code>	3/5-channel geometric attributes
<code>_directed_relation_hausdorff_distances</code>	asymmetric set distances

Keeping the shared message path centralized prevents three coordinate policies from drifting into three subtly different ETNN implementations.

17 What this implementation contributes

The ETNN architecture and its theoretical properties originate in the ETNN paper (Battiloro et al. 2025). The official NSAPH repository supplies the primary software reference (Battiloro et al. 2024). Within those boundaries, the TopoBench implementation contributes the following engineering design:

1. one consolidated backbone with explicit, nonautomatic coordinate semantics;
2. typed sparse neighborhood adaptation across multiple cell ranks;
3. placeholder-safe batching for complexes with missing ranks;
4. recursive structural-coordinate lifting through incidence matrices;
5. reconstruction of physical cell memberships from TopoBench incidences;
6. physical invariant generation and normalization per relation;
7. internal equivariant position updates with inter-layer geometry refresh;
8. a stable rank-indexed output contract for standard wrappers and readouts.

The result is not a replacement for the original formulation. It is a detailed translation of ETNN into a modular topological-learning framework, with policy boundaries that make the model’s geometric assumptions visible in code and configuration.

18 Conclusion

Implementing ETNN requires coordinating three forms of structure: cell rank, topological relation type, and geometry. The central implementation lesson is that geometry should not be inferred from attribute names or hidden fallback logic. It should be an explicit contract.

Once that contract is established, the implementation has a coherent shape. Sparse TopoBench relations determine who communicates. Coordinate policies determine which scalar attributes accompany those communications. Gated relation modules aggregate messages. Rank-specific residual modules update cell features. In physical mode, invariant scalars control relative-vector coordinate displacements, and changed geometry is recomputed before the next layer.

That separation preserves the original ETNN mechanism where physical geometry is meaningful while allowing the same rank-wise topological scaffold to be used, with clearly limited semantics, when geometry is absent or structural.

References

- Battiloro, Claudio, Ege Karaismailolu, Mauricio Tec, George Dasoulas, Michelle Audirac, and Francesca Dominici. 2024. *Topological Equivariant Networks: Official Implementation of E(n) Equivariant Topological Neural Networks*. V. commit 639e35e. Released. <https://github.com/NSAPH-Projects/topological-equivariant-networks>.
- Battiloro, Claudio, Ege Karaismailolu, Mauricio Tec, George Dasoulas, Michelle Audirac, and Francesca Dominici. 2025. “E(n) Equivariant Topological Neural Networks.” *The Thirteenth International Conference on Learning Representations*. <https://openreview.net/forum?id=Ax3uIiEBVR>.
- Dwivedi, Vijay Prakash, Chaitanya K. Joshi, Thomas Laurent, Yoshua Bengio, and Xavier Bresson. 2023. “Benchmarking Graph Neural Networks.” *Journal of Machine Learning Research* 24 (43): 1–48. <https://jmlr.org/papers/v24/22-0567.html>.
- Geometric Intelligence Lab. 2026. *TDL Challenge 2026: Bridging the Gap*. TopoBench documentation. <https://geometric-intelligence.github.io/topobench/tdl-challenge-2026/index.html>.
- Hajji, Mustafa, Ghada Zamzmi, Theodore Papamarkou, et al. 2022. “Topological Deep Learning: Going Beyond Graph Data.” *arXiv Preprint arXiv:2206.00606*, ahead of print. <https://doi.org/10.48550/arXiv.2206.00606>.
- Khanal, Gaurav. 2026. *TopoBench Coordinate-Policy ETNN Implementation*. V. commit a0bd4113. Released. https://github.com/grvkhnL/TopoBench/blob/a0bd4113f431efce8eb1ca38dcfe4837ab995779/topobench/nn/backbones/combinatorial/etnn_coordinate_policy.py.
- Ramakrishnan, Raghunathan, Pavlo O. Dral, Matthias Rupp, and O. Anatole von Lilienfeld. 2014. “Quantum Chemistry Structures and Properties of 134 Kilo Molecules.” *Scientific Data* 1: 140022. <https://doi.org/10.1038/sdata.2014.22>.
- Satorras, Víctor Garcia, Emiel Hoogeboom, and Max Welling. 2021. “E(n) Equivariant Graph Neural Networks.” *Proceedings of the 38th International Conference on Machine Learning* 139: 9323–32. <https://proceedings.mlr.press/v139/satorras21a.html>.
- Telyatnikov, Lev, Guillermo Bernardez, Marco Montagna, et al. 2025. “TopoBench: A Framework for Benchmarking Topological Deep Learning.” *Journal of Data-Centric Machine Learning Research*. <https://openreview.net/forum?id=07sTzyEVtY>.
- Van Langendonck, Louis, Guillermo Bernárdez, Nina Miolane, and Pere Barlet-Ros. 2026. “GraphUniverse: Synthetic Graph Generation for Evaluating Inductive Generalization.” *International Conference on Learning Representations*. <https://openreview.net/forum?id=jRWxvQnqUt>.

19 Appendix: reproducibility and attribution record

19.1 Primary implementation

```
1 Repository: grvkhnL/TopoBench
2 File: topobench/nn/backbones/combinatorial/etnn_coordinate_policy.py
3 Revision: a0bd4113f431efce8eb1ca38dcfe4837ab995779
4 License: MIT
```

19.2 Original architecture

```
1 Claudio Battiloro, Ege Karaismailoğlu, Mauricio Tec, George Dasoulas,  
2 Michelle Audirac, and Francesca Dominici.  
3 "E(n) Equivariant Topological Neural Networks."  
4 The Thirteenth International Conference on Learning Representations, 2025.  
5 https://openreview.net/forum?id=Ax3uIiEBVR
```

19.3 Reference software

```
1 Repository: NSAPH-Projects/topological-equivariant-networks  
2 Revision examined: 639e35e  
3 URL: https://github.com/NSAPH-Projects/topological-equivariant-networks
```

Repository behavior is cited as software behavior, not attributed to a paper equation unless the paper independently specifies it.

19.4 Validation revisions and evidence records

The validation studies were completed before the final report revision. Their exact code states and evidence records are listed below so that the reported numbers are not implicitly attributed to a later checkout. The changes from the QM9 validation revision to the report revision in the shared ETNN core and parity harness are editorial comments only; the executable statements are unchanged.

GraphUniverse coordinate-free and structural policies. TopoBench revision 776dcf4972974d17df36869f84e8f84302ec2f2c; evidence: two committed results.json files.

Reduced public physical path. TopoBench revision 16078dc35db1a621dd232e9f43549ba8088925e2; evidence: three archived configuration/metric pairs.

QM9 adapter and model-core parity. TopoBench revision 2286a7b0bd8dae02a7d596b2bc182b6f77b8010e; evidence: real-native-batch-model-parity.json.

Protocol-matched TopoBench ETNN core. TopoBench revision 2286a7b0bd8dae02a7d596b2bc182b6f77b8010e; evidence: training-record.json and epoch-metrics.csv.

Native NSAPH reproduction. Pinned NSAPH revision 639e35e; evidence: run-record.json and metrics.jsonl.

The SHA-256 digests of the records used for the tables and figures are:

17fab76c227d5ab581895d28fed5480257ceb51cc0e23ac673e0e11fef39d9d0	coordinate-free results.json
6f93989221d9c272336806afe47e510a919da0c3e010de26b11a1110635204aa	structural-coordinate results.json
660d79f220e02a461a73e3e5b4171eb5c3aef58496ae302135e4ce87a19a8761	model-core parity record
574432d1915cf19606fb2a2a3a3954ea4f6e76f702a4015128575aa3e2f1a220	native 1,000-epoch run record
66993a4ca71e4cb697d7585e9bb3ecee1bf12e57db0cc7527d3e3bc753532f8	native epoch metrics
7296eda38f2dee74f3d5e065a4d8e23091f08c074e47936341cc97d4cb81cf78	TopoBench core training record
da693aad89000bacb857632a4008dd0c9b0dea730ea38b42038d30b770509201	TopoBench core epoch metrics

For the reduced physical-path study, the archived configuration and metric digests are paired by variant:

```
1 fixed coordinates: hparams 01eec27b91f1d6de201fba7e2a2236c49f44a721abafba62a9d0b5f85cda6e9a
2 metrics fe90e9460dddb719e7e63c02eaa4d3ead2c8d5a0c1a85e2965c485f65b22e51f
3 dynamic coordinates: hparams b0c59fefaf1f5de25453ad948681a53c476402e399651b0dc6aa4e8c0e462b1a
4 metrics 9d7f487ada2ae420ba0afbc6dcd50e39db2d99bb5ad05b422c5a3379a9e6cdcb
5 full physical policy: hparams 41a17a75c5e8a890bbea3edc73aa4929af54f4925105ca3ae6b1cf2c935ca274
6 metrics 5c926539ac79dd28c5cc71a78d7e85adb71e820c29d56d7ee09700dec348e996
```